Tusher Bhomik

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Education

- **Bangladesh University of Engineering and Technology** — *January 2022 – June 2026, CGPA: 3.3164/4.00*
  Bachelor of Science in Computer Science & Engineering — Dhaka, Bangladesh

Relevant Coursework

- Machine Learning, Artificial Intelligence, Data Structures, Algorithms, Theory of Computation, Computer Architecture, Operating Systems, Compiler Design, Database Management Systems, Computer Networks, Software Engineering, Computer Graphics

Research

- **Age-Aware Neurological Gait Pattern Discovery from Biomechanical Signals: Multi-Phase Stroke Severity Analysis** — *2025 – 2026*
  Undergraduate Thesis (defended) | Supervisor: Mahmuda Naznin, Dept. of CSE, BUET — Dhaka, Bangladesh
  - An **age-normative framework** for uncovering neurological relations from biomechanical gait features using **machine learning with explainable AI**.
- **Reinforcement Learning for Reliable and Security-Aware LLM Agents (Ongoing)** — *2026 – Present*
  Supervisor: Md. Rizwan Parvez, Qatar Computing Research Institute — Doha, Qatar
  - Investigating reinforcement learning-based optimization to enhance the reliability and accuracy of function calling behavior in large language models for complex tool-use scenarios.
  - Exploring reinforcement learning techniques for developing security-aware agentic systems capable of secure reasoning, vulnerability analysis, and autonomous decision-making in software repositories.

Projects

- **HealthSync** — *2025*
  Spring Boot, React, PostgreSQL, Firebase — GitHub
  - Developed a comprehensive healthcare platform connecting patients and doctors with appointment scheduling, digital prescription management, and medical history tracking.
- **Arxiv Research Paper RAG System** — *2026*
  Python, SQL, SQLite, FastAPI, ChromaDB, Data Analysis — GitHub
  - Built an end-to-end **data pipeline** for ingesting, cleaning, and analyzing arXiv research paper data using Python and SQL.
  - Developed a **RAG-based API system** using FastAPI and ChromaDB for semantic retrieval and question-answering.
- **Evidence-Grounded Multilingual Image-Edit Evaluator** — *2026*
  Python, VLMs (Qwen, Gemma), Multilingual Benchmark
  - Built a multilingual benchmark (∼6K image-edit pairs from **Pico-Banana-400K**, English/Bangla/Hindi) for instruction-following image editing.
  - Designed an **evidence-grounded multi-label evaluator** fusing visual and textual features to diagnose **11 fine-grained failure categories** and benchmarked against open-source VLM judges (Qwen, Gemma).
- **Enhanced xv6-riscv Kernel** — *2024*
  C, Assembly, Makefile — GitHub
  - Implemented an **MLFQ scheduler** with aging and added **kernel-level threading** support.
  - Added custom system calls (`trace`, `sysinfo`) and a Bash autograder for testing.
- **Mini C Compiler** — *2024*
  CSE-310 Compiler Design, C, 8086 Assembly — GitHub
  - Built a **mini C compiler** from scratch in four phases: Symbol Table, Lexical Analysis, Semantic Analysis, and Intermediate Code Generation.
  - Generated **8086 assembly** as intermediate output to validate end-to-end compilation workflow.

Achievements

- **5th Place — Tech Career Hunt 2.0**, organized by Shohoj Coding. — *May 2026*
- **Internship Selection at Save the Children —** AI/ML Track Finalist in their Hackathon.

Skills

- **Research Interests:** LLMs, Reinforcement Learning, Trustworthy AI, AI Security, Software Security, Systems Security, Operating Systems, Human-Computer Interaction, Cyber-Physical Systems
- **Programming:** Python, C++, C, Java, JavaScript | **Databases:** MySQL, PostgreSQL, MongoDB
- **ML/AI:** PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, OpenCV, LLMs, vLLM
- **Tools:** Git, Docker, Linux | **Frontend:** React, Next.js, Material-UI

Additional Information

Languages: English, Bengali. **Interests:** Photography, Traveling, Badminton, Cycling.

References

- **Prof. Dr. Mahmuda Naznin** — Thesis Supervisor
  Department of CSE, Bangladesh University of Engineering and Technology
  Email: mahmudanaznin@gmail.com